NYU

ROB-UY 3303

Robot Motion & Planning

Lecture 01A - Course Introduction



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Agenda

1. **Motivation**
2. **People**
3. **Course Objectives**
4. **Lectures**
5. **Assignments**
6. **Projects**
7. **Grading**

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Lead Instructor

- **Christopher Clark**
  - Email: c.clark@nyu.edu
  - Office Hours - 3:30-5:00am Mondays



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TEACHING ASSISTANTS

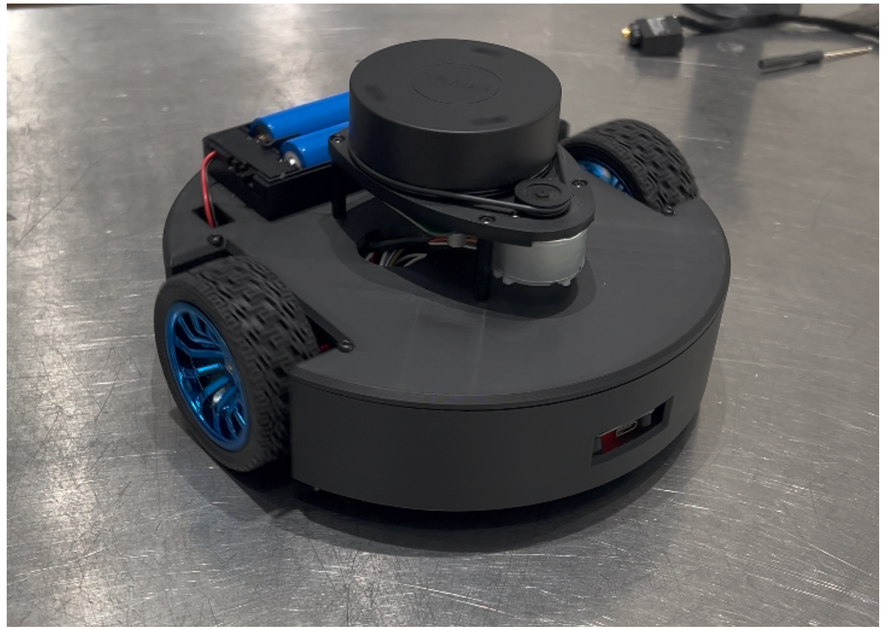
- **Matteo Piras** - mp7471@nyu.edu
  - Office Hours Location: Jacobs Hall, 339
  - Office Hours: Thursdays, 6:00pm - 9:00pm
- **Kamil Imam** - mki4908@nyu.edu
  - Office Hours Location: Jacobs Hall, 339
  - Office Hours: Fridays, 3:00pm - 6:00pm

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Course Overview

This course covers the concepts, techniques, algorithms, and state-of-the-art approaches for robot motion planning, and task planning. We first understand the robot control architecture where robots invoke “planning” to enable autonomous navigation. Next, we will model the motion of a robot, understand how to localize it, and explore a large number of planning algorithms that will enable state to state autonomous navigation. Real robot systems will be used as a teaching vessel.



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LECTURE SCHEDULE

- Week 1
  - Sep. 02, 2026 Introduction, History of MP, the MP Problem
- Week 2
  - Sep. 07, 2026 Labor Day - no class
  - Sep. 09, 2026 Coordinate Frames, Kinematics, Probability
- Week 3
  - Sep. 14, 2026 Configuration Space of Different Robots
  - Sep. 16, 2026 The Differential Drive Robot
- Week 4
  - Sep. 21, 2026 Trajectory Tracking & Control
  - Sep. 23, 2026 Tree Search Algorithms I
- Week 5
  - Sep. 28, 2026 Tree Search Algorithms II
  - Sep. 30, 2026 Sampling Based Algorithms I
- Week 6
  - Oct. 05, 2026 Sampling Based Algorithms I
  - Oct. 07, 2026 POMDP Planning I
- Week 7
  - Oct. 12, 2026 Fall break - No class
  - Oct. 14, 2026 POMDP Planning II
- Week 8
  - Oct. 19, 2026 Active Perception
  - Oct. 21, 2026 Multi-Robot MP
- Week 8
  - Oct. 19, 2026 Active Perception
  - Oct. 21, 2026 Multi-Robot MP
- Week 9
  - Oct. 26, 2026 Optimization Based Planning I
  - Oct. 28, 2026 Optimization Based Planning II
- Week 10
  - Nov. 02, 2026 Midterm Exam Review
  - Nov. 04, 2026 Midterm Exam
- Week 11
  - Nov. 09, 2026 APF Olympics
  - Nov. 11, 2026 Novel Robot MP
- Week 12
  - Nov. 16, 2026 ML Based MP I
  - Nov. 18, 2026 ML Based MP II
- Week 13
  - Nov. 23, 2026 Project Kickoff
  - Nov. 25, 2026 Project Proposals
- Week 14
  - Nov. 30, 2026 Project Lit. Review I
  - Dec. 02, 2026 Project Lit. Review II
- Week 15
  - Dec. 07, 2026 Project working session
  - Dec. 09, 2026 Project working session
- Week 16
  - Dec. 14, 2026 Project Final Presentations

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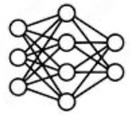
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ASSIGNMENT SCHEDULE

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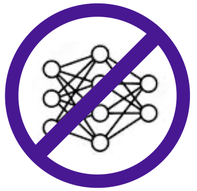
For assignments 2, 4, 5, and 6, we will divide student pairs into two groups - group A and group B. For these assignments, one of the two groups will complete their assignment work in the ZeroAI lab, and the other group will complete their work whereever they choose - using Generative AI tools. For this later group, instructions will be provided on how to use the GenAI tools.

Group A



Directed use of GenAI Tools

Group B



Absolutely no use of GenAI Tools

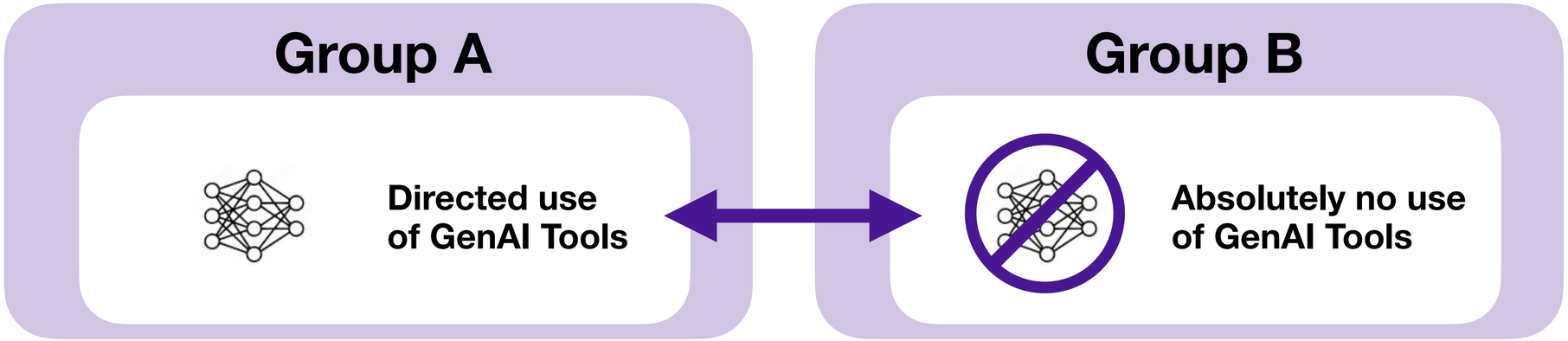
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▪ Sep. 09, 2026	Assignment 1 - Robot Building	Due Sep. 13
▪ Sep. 14, 2026	Assignment 2 - Robot Motion Modeling	Due Sep. 27
▪ Sep. 28, 2026	Assignment 3 - Trajectory Following	Due Oct. 04
▪ Oct. 05, 2026	Assignment 4 - Tree Search MP	Due Oct. 18
▪ Oct. 19, 2026	Assignment 5 - Sampling Based MP	Due Nov. 01
▪ Nov. 09, 2026	Assignment 6 - Multi-Robot MP	Due Nov. 22

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PROJECT OVERVIEW

The goal of the final project is to allow students to explore and learn a novel approach to robot motion planning. The project should be an extension of work from class labs and assignments, but will leverage the pupper robot system hardware. Project requirements include:

- Students must deploy on a real robot
- Students must have a performance metric
- Students must work in groups of 2 people, (same groups as lab groups)

DELIVERABLES & GRADING

The following deliverables will make up your project's grade. Note that there is a rubric attached to each deliverable that provides guidance on expectations and grading. Note that your peer evaluation scores affect grades in two ways: 1) the evaluation you give a presenter affects the presenter's grade, 2) the evaluation you give a presenter affects YOUR grade. This is used to assess how much you are learning from other student's presenations.

Project Proposal - 10%

- 1-page written project proposal

Literature Review - 20%

- Live presentation of paper - 10%
- Peer evaluations of video presentation (same rubric as above) - 10%

Final Presentation - 70%

- Live presentation of final project - 60%
- Peer evaluations of presentation (same rubric as above) - 10%

Grading

1. Projects – 35%
2. Assignments - 35%
3. Midterm Exam - 30%

Grading Late Work Policy

- You will have 4 late days to use on assignments throughout the semester.
  - NO QUESTIONS ASKED
  - NO NEED TO REPORT
  - You can divide up your “4” late “days” however you want (all in one assignment or spread across many. This is NOT 96 hours to split up.

- No late work is accepted for project presentations, unless major emergencies.

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Course Website



https://sites.google.com/nyu.edu/rob-uy3303/